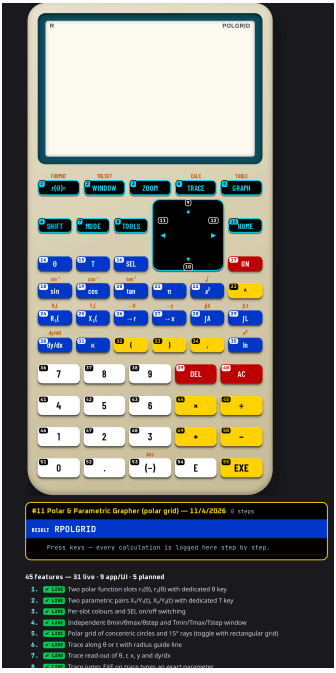
U-45 CALCULATORS

User Manual

Calculator #11 Polar & Parametric Grapher (polar grid)

Graphing – colour



Complete kit: functions · numbered key map · worked examples · tricks & hacks · shortcuts · common mistakes · practice drills

About this manual

This manual belongs to the **U-45 CALCULATORS** app and describes calculator **#11 — Polar & Parametric Grapher (polar grid)** exactly as it behaves in the app. Every worked example and every drill answer in this book was run on the app's own keys by an automated checker before printing; only results that matched were kept.

How key presses are written

Keys are written as a chip showing the **key number** and its face, for example [#1 r(θ)=]. Numbers follow the numbering printed on the keys (switch it on with "# On"). A shifted function is written as two chips: the shift key, then the key that carries the function.

Entry style of this model

polar r(θ) and parametric X(t),Y(t) graphing with polar grid.

Chapters

1. Getting started
2. Features
3. Feature guide (2b)
4. Numbered key map
5. Key-by-key reference
6. Worked examples
7. Tricks & hacks
8. Keyboard shortcuts
9. Common mistakes
10. Practice drills
11. Appendix A — Drill answers

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14. Appendix D — Model notes & limits
15. Appendix E — Quick-reference card

1 · Getting started

Open U-45 CALCULATORS, choose **#11** from the home screen (or the jump list at the top), and the calculator appears with its numbered, colour-coded keys. Click or tap keys, or type on a keyboard.

Numbers on every key

Every key carries a small number in the same reading order as this manual (left → right, top → bottom; the navigation pad counts ▲ ▼ ◀ ▶ then its centre). Use "# On / # Off" in the bar above the calculator to show or hide them — the setting is remembered.

High-contrast colours

Keys are coloured by job: white = digits, yellow = operators/execute, blue = functions, green = memory/variables, red = clear/delete/power, black with cyan = shift/modes/menus/navigation. Press "Classic" to return to the original model colours and "Hi-Contrast" to come back.

Three typefaces

Display text uses the U45 Display face, key legends use U45 Keys (a condensed face that stays readable on small keys) and key numbers / helper text use U45 Mono. All three are bundled inside the app, so they work offline.

Copy the display

The Copy button copies exactly what the display shows (indicators that are switched off are skipped) so you can paste a result into a message or spreadsheet.

Fit or 1:1

Fit scales the calculator to your screen; 1:1 shows it at its design size. On a phone, Fit is the comfortable choice.

Reset

Reset rebuilds this calculator from scratch: display, memory, stack and modes return to their start-up state. Use it before each worked example in this manual.

Keyboard

On a computer you can type: digits, + − * / ^ ( ), Enter or = to execute, Backspace to delete, Esc to clear, arrows to move. Letters reach common functions (see the Keyboard shortcuts chapter).

Tooltips

Hover over (or long-press) a key to see "#number name".

Tools rail (v1.4)

A cyan icon rail on the right of every calculator opens 15 panels: Currency (166 currencies, rates auto-updated from open.er-api.com or Frankfurter when online and cached with a date; the bundled dated snapshot is used offline), Units (exact SI factors), Health (published formulas, not medical advice), Simple, Complex, Loans, Construction, Mechanical and Engineering (30+ numbered tools each), Calendar generator, Customize (26 options, saved per calculator), Settings (31 options), History, Hide and View. On narrow phones the rail shows icons only.

History (v1.4)

Every calculation is logged per calculator with a date and time. Open History on the rail to search, re-use (types the input back into the calculator), copy, delete one, select several and delete, or delete all (asks for confirmation). History is stored encrypted on this device (AES-GCM); exporting it needs the passcode.

Camera Solve (v1.4)

The green CS button between Restore and Social reads a problem from the camera, a photo, a PDF or a text file. Recognition (Tesseract OCR, English) runs offline on the device; an online language download is only used if you switch it on in Settings. Check and edit the recognised text, then Solve shows the steps offline. OCR is not available when the app is opened as a plain file:// page in a browser.

Layouts and view modes (v1.4)

The layout selector between Fit and # offers Auto plus 12 layouts: 3D classic, 3D high-contrast, Classic, Pocket, Compact, Expanded, Desktop wide, Split, One-hand left, One-hand right, Large print and Minimal. Auto picks one for your screen and your choice is remembered per calculator. The View panel hides the top bars, the tool bar, the rail or everything (full screen); the round yellow ⤢ button at bottom-left (or Esc) always restores the normal view.

Hide items (v1.4)

The Hide panel lists every button, rail icon and section of this calculator as tick boxes. Hiding is instant; unhiding asks for the app passcode.

Image protection (v1.4)

Right-click, drag and long-press saving of images is blocked inside the app. Saving or sharing an image export (PNG, JPG, QR, calendar), printing a calendar, and downloading or sharing the app itself ask for the passcode, and exported images carry a U-45 watermark. Honest note: this cannot stop a screenshot or screen recording made by the operating system or another device.

Works offline

U-45 CALCULATORS is an offline app: after the first load everything — including fonts — comes from the device.

Colour groups on this calculator

Colour	Group	Keys on this model
num	Digits & number entry	#36, #37, #38, #41, #42, #43, #46, #47, #48, #51, #52, #53, #54
op	Operators & execute	#23, #32, #33, #34, #44, #45, #49, #50, #55
fn	Functions	#14, #15, #16, #18, #19, #20, #21, #22, #24, #25, #26, #27, #28, #29, #30, #31, #35
mem	Memory, variables & registers	—
clr	Clear, delete & power	#17, #39, #40

mod	**Shift, modes, menus & navigation**	**#1, #2, #3, #4, #5, #6, #7, #8, #9, #10, #11, #12, #13**

2 · Features

45 features: 31 live on the keys, 9 app/UI features, 5 planned (documented but not yet working — they are clearly marked and never used in examples).

1. LIVE **Two polar function slots $r_1(\theta)$, $r_2(\theta)$ with dedicated θ key**
2. LIVE **Two parametric pairs $X_1/Y_1(t)$, $X_2/Y_2(t)$ with dedicated T key**
3. LIVE **Per-slot colours and SEL on/off switching**
4. LIVE **Independent θmin/θmax/θstep and Tmin/Tmax/Tstep window**
5. LIVE **Polar grid of concentric circles and 15° rays (toggle with rectangular grid)**
6. LIVE **Trace along θ or t with radius guide line**
7. LIVE **Trace read-out of θ, r, x, y and dy/dx**
8. LIVE **Trace jump: EXE on trace types an exact parameter**
9. LIVE **Switch traced curve with ▲ ▼**
10. LIVE **Tool: evaluate $r_1(\theta)$ with tangent line overlay**
11. LIVE **Tool: evaluate $X_1,Y_1(t)$ with dx/dt, dy/dt and tangent**
12. LIVE **Tool: polar area $\frac{1}{2}\int r^2 d\theta$ with shaded sector**
13. LIVE **Tool: polar arc length with highlighted arc**
14. LIVE **Tool: parametric arc length with highlighted path**
15. LIVE **Tool: enclosed area by Green's theorem**
16. LIVE **Tool: curvature κ(t) and radius of curvature**
17. LIVE **Tool: $r_1 = r_2$ intersection finder (marked points)**
18. LIVE **Tool: farthest point max|r| by golden-section search**
19. LIVE **Rect → polar and polar → rect point converters**

20. LIVE **Zoom menu: in, out, standard, square, integer, decimal**
21. LIVE **Zoom trig (θ, t reset to 0…2π)**
22. LIVE **Zoom fit to visible curves**
23. LIVE **Sample-dot display to show θ/t spacing**
24. LIVE **Table of θ/t with r_1, r_2, X_1, Y_1 columns**
25. LIVE **Home functions R1() R2() X1() Y1() X2() Y2() evaluate slots**
26. LIVE **Home functions tor(), toth(), torx(), tory() for coordinate conversion**
27. LIVE **Home integrals parea(), plen(), tlen(), garea()**
28. LIVE **Home slopes pslope(θ) and tslope(t), curvature curv(t)**
29. LIVE **Radian-native graphing; home angle mode switchable**
30. LIVE **Pan the view with the cursor cross when not tracing**
31. LIVE **Window form with 12 fields**
32. APP/UI **Every key numbered 1…N in reading order — matches the key map in the user manual**
33. APP/UI **High-contrast 6-colour key groups (digits, operators, functions, memory, clear, modes) with a "Classic" toggle**
34. APP/UI **Three bundled offline fonts: Oxanium (display/titles), Barlow Condensed (key faces), JetBrains Mono (key numbers & helpers)**
35. APP/UI **Show / hide key numbers with one button (remembered between sessions)**
36. APP/UI **Copy the current display text to the clipboard**
37. APP/UI **Physical-keyboard shortcuts: digits, + − * / ^ ( ) . Enter Backspace Esc arrows, s c t l n r ! % e p q i**
38. APP/UI **Fit / 1:1 zoom that never pushes keys off a phone screen**
39. APP/UI **Reference photo viewer and tool-tip name on every key**

40. APP/UI **Reset = fresh power-on state; works fully offline (PWA cache / Android APK)**
41. PLANNED **Graph image export (PNG)**
42. PLANNED **Shareable function list link**
43. PLANNED **Dark screen theme**
44. PLANNED **Speech read-out of values**
45. PLANNED **Exam mode**

2b · Feature guide

Calculator #11 is a polar r(θ) and parametric X(t),Y(t) graphing with polar grid. Each card below is one of its 31 working features, with the keys that reach it and, where one exists, a verified example.

F1 Two polar function slots $r_1(\theta)$, $r_2(\theta)$ with dedicated θ key

Where: #1 r(θ)= Edit polar and parametric slots; #4 TRACE TRACE — move along a graph; #6 SHIFT SHIFT / 2nd — the next key uses its upper (shift-coloured) function

See it in action: Rectangular to polar radius — *tor(3,4)*= 5; Evaluate the polar slot $r_1 = 2\cos\theta$ at θ = 0 — *R1(0)*= 2

Try it: Press Reset, use the key(s) above, and write the display here:

_____

Your notes:

_____

_____

F2 Two parametric pairs $X_1/Y_1(t)$, $X_2/Y_2(t)$ with dedicated T key

Where: #1 r(θ)= Edit polar and parametric slots; #4 TRACE TRACE — move along a graph; #8 TOOLS Polar/parametric tools

See it in action: Parametric $X_1(t) = 3\cos t$ at t = 0 — *X1(0)*= 3

Try it: Press Reset, use the key(s) above, and write the display here:

_____

Your notes:

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_____

F3 Per-slot colours and SEL on/off switching

Where: #16 SEL Turn the current slot on/off; #24 $R_1($ Evaluate polar slot r_1

See it in action: Evaluate the polar slot $r_1 = 2\cos\theta$ at $\theta = 0$ — *R1(0)*= 2

Try it: Press Reset, use the key(s) above, and write the display here:

_____

Your notes:

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_____

F4 Independent θmin/θmax/θstep and Tmin/Tmax/Tstep window

Where: #2 WINDOW WINDOW — set the graph view

Try it: Press Reset, use the key(s) above, and write the display here:

_____

Your notes:

_____

_____

F5 Polar grid of concentric circles and 15° rays (toggle with rectangular grid)

Where: #1 r(θ)= Edit polar and parametric slots; #4 TRACE TRACE — move along a graph; #8 TOOLS Polar/parametric tools

See it in action: Rectangular to polar radius — *tor(3,4)*= 5; Evaluate the polar slot $r_1 = 2\cos\theta$ at θ = 0 — *R1(0)*= 2

Try it: Press Reset, use the key(s) above, and write the display here:

_____

Your notes:

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_____

F6 Trace along θ or t with radius guide line

Where: #4 TRACE TRACE — move along a graph; #9 ▲ Cursor ▲ — up / previous history line / previous item; #10 ▼ Cursor ▼ — down / next history line / next item

See it in action: Rectangular to polar radius — *tor(3,4)*= 5

Try it: Press Reset, use the key(s) above, and write the display here:

_____

Your notes:

_____

_____

F7 Trace read-out of θ, r, x, y and dy/dx

Where: #4 TRACE TRACE — move along a graph

Try it: Press Reset, use the key(s) above, and write the display here:

_____

Your notes:

_____

_____

F8 Trace jump: EXE on trace types an exact parameter

Where: #4 TRACE TRACE — move along a graph; #15 T Parameter variable; #21 π π (3.14159265…) as an exact constant

Try it: Press Reset, use the key(s) above, and write the display here:

_____

Your notes:

_____

_____

F9 Switch traced curve with ▲ ▼

Where: from the TOOLS / menu screens of this model (see chapter 4 for the menu key)

Try it: Press Reset, use the key(s) above, and write the display here:

_____

Your notes:

_____

_____

F10 Tool: evaluate $r_1(\theta)$ with tangent line overlay

Where: #55 EXE Execute / = — evaluate the line and show the result (EXE / ENTER / =); #9 ▲ Cursor ▲ — up / previous history line / previous item; #10 ▼ Cursor ▼ — down / next history line / next item

See it in action: Evaluate the polar slot $r_1 = 2\cos\theta$ at $\theta = 0$ — *R1(0)*= 2

Try it: Press Reset, use the key(s) above, and write the display here:

_____

Your notes:

_____

_____

F11 Tool: evaluate $X_1,Y_1(t)$ with dx/dt, dy/dt and tangent

Where: #20 tan Tangent (current angle unit); #24 R_1(Evaluate polar slot r_1; #25 X_1(Evaluate $X_1(t)$

See it in action: Evaluate the polar slot $r_1 = 2\cos\theta$ at $\theta = 0$ — *R1(0)*= 2

Try it: Press Reset, use the key(s) above, and write the display here:

_____

Your notes:

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_____

F12 Tool: polar area ½∫r²dθ with shaded sector

Where: #28 ∫A Polar area of r_1; #1 r(θ)= Edit polar and parametric slots; #4 TRACE TRACE — move along a graph

See it in action: Rectangular to polar radius — *tor(3,4)*= 5; Evaluate the polar slot $r_1 = 2\cos\theta$ at θ = 0 — *R1(0)*= 2

Try it: Press Reset, use the key(s) above, and write the display here:

_____

Your notes:

_____

_____

F13 Tool: polar arc length with highlighted arc

Where: #29 ∫L Polar arc length of r_1; #1 r(θ)= Edit polar and parametric slots; #4 TRACE TRACE — move along a graph

See it in action: Rectangular to polar radius — *tor(3,4)*= 5; Evaluate the polar slot $r_1 = 2\cos\theta$ at θ = 0 — *R1(0)*= 2

Try it: Press Reset, use the key(s) above, and write the display here:

_____

Your notes:

_____

_____

F14 Tool: parametric arc length with highlighted path

Where: #29 ∫L Polar arc length of r_1; #1 r(θ)= Edit polar and parametric slots; #4 TRACE TRACE — move along a graph

See it in action: Arc length of r = 2cos θ for θ in [0, π] — *plen(0,π)*= 6.28318531; Parametric $X_1(t)$ = 3cos t at t = 0 — *X1(0)*= 3

Try it: Press Reset, use the key(s) above, and write the display here:

_____

Your notes:

_____

_____

F15 Tool: enclosed area by Green's theorem

Where: #28 ∫A Polar area of r_1

See it in action: Area of the circle r = 2cos θ (θ from 0 to π) — *parea(0,π)*= 3.14159265; Area enclosed by the ellipse (Green's theorem) — *garea(0,2π)*= 18.84955592

Try it: Press Reset, use the key(s) above, and write the display here:

_____

Your notes:

_____

_____

F16 Tool: curvature κ(t) and radius of curvature

Where: #31 κ Curvature of X_1Y_1; #26 →r Radius from x,y

See it in action: Rectangular to polar radius — *tor(3,4)*= 5

Try it: Press Reset, use the key(s) above, and write the display here:

_____

Your notes:

_____

_____

F17 Tool: $r_1 = r_2$ intersection finder (marked points)

Where: from the TOOLS / menu screens of this model (see chapter 4 for the menu key)

Try it: Press Reset, use the key(s) above, and write the display here:

_____

Your notes:

_____

_____

F18 Tool: farthest point max|r| by golden-section search

Where: #52 . Decimal point. Pressing it first gives "0."; a second point in the same number is ignored

Try it: Press Reset, use the key(s) above, and write the display here:

_____

Your notes:

_____

_____

F19 Rect → polar and polar → rect point converters

Where: #1 r(θ)= Edit polar and parametric slots; #4 TRACE TRACE — move along a graph; #8 TOOLS Polar/parametric tools

See it in action: Rectangular to polar radius — *tor(3,4)*= 5; Evaluate the polar slot $r_1 = 2\cos\theta$ at θ = 0 — *R1(0)*= 2

Try it: Press Reset, use the key(s) above, and write the display here:

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Your notes:

_____

_____

F20 Zoom menu: in, out, standard, square, integer, decimal

Where: #3 ZOOM Zoom menu; #22 x^2 Square x^2; #52 . Decimal point. Pressing it first gives "0."; a second point in the same number is ignored

Try it: Press Reset, use the key(s) above, and write the display here:

_____

Your notes:

_____

_____

F21 Zoom trig (θ, t reset to 0…2π)

Where: #3 ZOOM Zoom menu

Try it: Press Reset, use the key(s) above, and write the display here:

_____

Your notes:

_____

_____

F22 Zoom fit to visible curves

Where: #3 ZOOM Zoom menu

Try it: Press Reset, use the key(s) above, and write the display here:

_____

Your notes:

_____

_____

F23 Sample-dot display to show θ/t spacing

Where: #7 MODE SETUP — angle unit, number format, display options

Try it: Press Reset, use the key(s) above, and write the display here:

_____

Your notes:

_____

_____

F24 Table of θ/t with r_1, r_2, X_1, Y_1 columns

Where: #2 WINDOW WINDOW — set the graph view; #5 GRAPH GRAPH — draw the graph

Try it: Press Reset, use the key(s) above, and write the display here:

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Your notes:

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F25 Home functions R1() R2() X1() Y1() X2() Y2() evaluate slots

Where: #1 r(θ)= Edit polar and parametric slots; #24 R_1(Evaluate polar slot r_1; #25 X_1(Evaluate $X_1(t)$

See it in action: Evaluate the polar slot $r_1 = 2\cos\theta$ at $\theta = 0$ — *R1(0)*= 2

Try it: Press Reset, use the key(s) above, and write the display here:

_____

Your notes:

_____

_____

F26 Home functions tor(), toth(), torx(), tory() for coordinate conversion

Where: from the TOOLS / menu screens of this model (see chapter 4 for the menu key)

See it in action: Rectangular to polar radius — *tor(3,4)*= 5; Polar to rectangular x for r = 2, θ = π/3 — *torx(2,π/3)*= 1

Try it: Press Reset, use the key(s) above, and write the display here:

_____

Your notes:

_____

_____

F27 Home integrals parea(), plen(), tlen(), garea()

Where: from the TOOLS / menu screens of this model (see chapter 4 for the menu key)

See it in action: Area of the circle r = 2cos θ (θ from 0 to π) — *parea(0,π)*= 3.14159265; Arc length of r = 2cos θ for θ in [0, π] — *plen(0,π)*= 6.28318531

Try it: Press Reset, use the key(s) above, and write the display here:

_____

Your notes:

_____

_____

F28 Home slopes pslope(θ) and tslope(t), curvature curv(t)

Where: #31 κ Curvature of X_1Y_1

See it in action: Slope of the ellipse X=3cos t, Y=2sin t at t = π/4 — *tslope(π/4)*= -0.66666667

Try it: Press Reset, use the key(s) above, and write the display here:

_____

Your notes:

_____

_____

F29 Radian-native graphing; home angle mode switchable

Where: #7 MODE SETUP — angle unit, number format, display options; #14 θ Polar angle variable; #18 sin Sine (uses the current angle unit)

Try it: Press Reset, use the key(s) above, and write the display here:

_____

Your notes:

_____

_____

F30 Pan the view with the cursor cross when not tracing

Where: #2 WINDOW WINDOW — set the graph view; #9 ▲ Cursor ▲ — up / previous history line / previous item; #10 ▼ Cursor ▼ — down / next history line / next item

Try it: Press Reset, use the key(s) above, and write the display here:

_____

Your notes:

_____

_____

F31 Window form with 12 fields

Where: #2 WINDOW WINDOW — set the graph view

Try it: Press Reset, use the key(s) above, and write the display here:

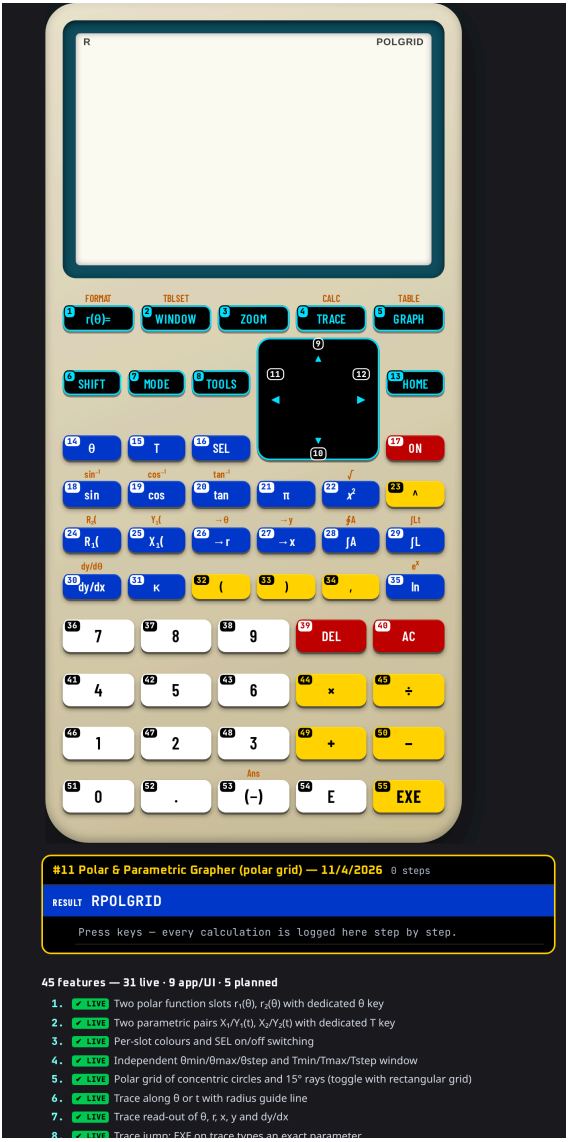
_____

Your notes:

_____

_____

3 · Numbered key map



Calculator #11 with key numbers shown (Hi-Contrast colours).

Key table

“Shift” = 2nd / f / SHIFT layer; “Alpha” = ALPHA layer; “3rd” = 3rd / g layer. A dot means the layer is empty on that key.

#	Face	Shift	Alpha	3rd	Group
1	r(θ)=	FORMAT	·	·	mod
2	WINDOW	TBLSET	·	·	mod
3	ZOOM	·	·	·	mod
4	TRACE	CALC	·	·	mod
5	GRAPH	TABLE	·	·	mod
6	SHIFT	·	·	·	mod
7	MODE	·	·	·	mod
8	TOOLS	·	·	·	mod
9	▲	·	·	·	mod
10	▼	·	·	·	mod
11	◀	·	·	·	mod
12	▶	·	·	·	mod
13	HOME	·	·	·	mod
14	θ	·	·	·	fn
15	T	·	·	·	fn
16	SEL	·	·	·	fn
17	ON	·	·	·	clr
18	sin	$\sin^{-1}$	·	·	fn
19	cos	$\cos^{-1}$	·	·	fn
20	tan	$\tan^{-1}$	·	·	fn
21	π	·	·	·	fn
22	x^2	√	·	·	fn
23	^	·	·	·	op

#	Face	Shift	Alpha	3rd	Group
24	$R_1($	$R_2($	·	·	fn
25	$X_1($	$Y_1($	·	·	fn
26	→r	→θ	·	·	fn
27	→x	→y	·	·	fn
28	∫A	∮A	·	·	fn
29	∫L	∫Lt	·	·	fn
30	dy/dx	dy/dθ	·	·	fn
31	K	·	·	·	fn
32	(	·	·	·	op
33	)	·	·	·	op
34	,	·	·	·	op
35	ln	ex	·	·	fn
36	7	·	·	·	num
37	8	·	·	·	num
38	9	·	·	·	num
39	DEL	·	·	·	clr
40	AC	·	·	·	clr
41	4	·	·	·	num
42	5	·	·	·	num
43	6	·	·	·	num
44	×	·	·	·	op
45	÷	·	·	·	op
46	1	·	·	·	num
47	2	·	·	·	num
48	3	·	·	·	num
49	+	·	·	·	op
50	−	·	·	·	op
51	0	·	·	·	num

#	Face	Shift	Alpha	3rd	Group
52	.	·	·	·	num
53	(–)	Ans	·	·	num
54	E	·	·	·	num
55	EXE	·	·	·	op

4 · Key-by-key reference

Every key on calculator #11, in number order, with every function it carries in this app.

#1 r(θ)= `Shift, modes, menus & navigation`

Primary: Edit polar and parametric slots

shifted (2nd / f / SHIFT layer) (press #6 first): FORMAT — Toggle polar / rectangular grid

#2 WINDOW `Shift, modes, menus & navigation`

Primary: WINDOW — set the graph view

shifted (2nd / f / SHIFT layer) (press #6 first): TBLSET — TBLSET — table start and step

#3 ZOOM `Shift, modes, menus & navigation`

Primary: Zoom menu

#4 TRACE `Shift, modes, menus & navigation`

Primary: TRACE — move along a graph

shifted (2nd / f / SHIFT layer) (press #6 first): CALC — Polar/parametric tools

#5 GRAPH `Shift, modes, menus & navigation`

Primary: GRAPH — draw the graph

shifted (2nd / f / SHIFT layer) (press #6 first): TABLE — Table of values

#6 SHIFT `Shift, modes, menus & navigation`

Primary: SHIFT / 2nd — the next key uses its upper (shift-coloured) function

#7 MODE `Shift, modes, menus & navigation`

Primary: SETUP — angle unit, number format, display options

#8 TOOLS `Shift, modes, menus & navigation`

Primary: Polar/parametric tools

#9 ▲ `Shift, modes, menus & navigation`

Primary: Cursor ▲ — up / previous history line / previous item

#10 ▼ `Shift, modes, menus & navigation`

Primary: Cursor ▼ — down / next history line / next item

#11 ◀ `Shift, modes, menus & navigation`

Primary: Cursor ◀ — move left in the line

#12 ▶ `Shift, modes, menus & navigation`

Primary: Cursor ▶ — move right in the line

#13 HOME `Shift, modes, menus & navigation`

Primary: Calculation screen

#14 θ `Functions`

Primary: Polar angle variable

#15 T `Functions`

Primary: Parameter variable

#16 SEL `Functions`

Primary: Turn the current slot on/off

#17 ON `Clear, delete & power`

Primary: ON — power on

#18 sin `Functions`

Primary: Sine (uses the current angle unit)

shifted (2nd / f / SHIFT layer) (press #6 first): $\sin^{-1}$ — Inverse sine $\sin^{-1}$

Keyboard: S

#19 cos Functions

Primary: Cosine (current angle unit)

shifted (2nd / f / SHIFT layer) (press #6 first): $\cos^{-1}$ — Inverse cosine $\cos^{-1}$

Keyboard: c

#20 tan Functions

Primary: Tangent (current angle unit)

shifted (2nd / f / SHIFT layer) (press #6 first): $\tan^{-1}$ — Inverse tangent $\tan^{-1}$

Keyboard: t

#21 π Functions

Primary: π (3.14159265…) as an exact constant

Keyboard: p

#22 x^2 Functions

Primary: Square x^2

shifted (2nd / f / SHIFT layer) (press #6 first): $\sqrt{}$ — Square root $\sqrt{}$

Keyboard: q

#23 ^ Operators & execute

Primary: Power x^y / ^ — raise to any power. Chains right-to-left (2^3^2 = 2^9 = 512)

Keyboard: ^

#24 $R_1($ Functions

Primary: Evaluate polar slot r_1

shifted (2nd / f / SHIFT layer) (press #6 first): $R_2($ — Evaluate r_2

#25 $X_1($ Functions

Primary: Evaluate $X_1(t)$

shifted (2nd / f / SHIFT layer) (press #6 first): $Y_1($ — Evaluate $Y_1(t)$

#26 → r Functions

Primary: Radius from x,y

shifted (2nd / f / SHIFT layer) (press #6 first): → θ — Angle from x,y

#27 → x Functions

Primary: x from r,θ

shifted (2nd / f / SHIFT layer) (press #6 first): → y — y from r,θ

#28 ∫A Functions

Primary: Polar area of r_1

shifted (2nd / f / SHIFT layer) (press #6 first): ∮A — Parametric enclosed area

#29 ∫L Functions

Primary: Polar arc length of r_1

shifted (2nd / f / SHIFT layer) (press #6 first): ∫Lt — Parametric arc length

#30 dy/dx Functions

Primary: Parametric slope

shifted (2nd / f / SHIFT layer) (press #6 first): dy/dθ — Polar slope

#31 K Functions

Primary: Curvature of X_1Y_1

#32 (Operators & execute

Primary: Opening bracket (

Keyboard: (

#33) Operators & execute

Primary: Closing bracket). Closing brackets at the end of a line may be omitted

Keyboard:)

#34 , Operators & execute

Primary: Comma / argument separator (between function arguments, list items or coordinates)

Keyboard: ,

#35 ln Functions

Primary: Natural logarithm ln

shifted (2nd / f / SHIFT layer) (press #6 first): ex — e^x — exponential

Keyboard: n

#36 7 Digits & number entry

Primary: Digit 7

Keyboard: 7

#37 8 Digits & number entry

Primary: Digit 8

Keyboard: 8

#38 9 Digits & number entry

Primary: Digit 9

Keyboard: 9

#39 DEL Clear, delete & power

Primary: DEL — delete the character at the cursor

Keyboard: Backspace Delete

#40 AC `Clear, delete & power`

Primary: AC — clear all of the current calculation

Keyboard: Escape

#41 4 `Digits & number entry`

Primary: Digit 4

Keyboard: 4

#42 5 `Digits & number entry`

Primary: Digit 5

Keyboard: 5

#43 6 `Digits & number entry`

Primary: Digit 6

Keyboard: 6

#44 × `Operators & execute`

Primary: Multiplication ×

Keyboard: * x

#45 ÷ `Operators & execute`

Primary: Division ÷

Keyboard: /

#46 1 Digits & number entry

Primary: Digit 1

Keyboard: 1

#47 2 Digits & number entry

Primary: Digit 2

Keyboard: 2

#48 3 Digits & number entry

Primary: Digit 3

Keyboard: 3

#49 + Operators & execute

Primary: Addition +

Keyboard: +

#50 – Operators & execute

Primary: Subtraction –

Keyboard: -

#51 0 Digits & number entry

Primary: Digit 0

Keyboard: 0

#52 . Digits & number entry

Primary: Decimal point. Pressing it first gives "0."; a second point in the same number is ignored

Keyboard: .

#53 (–) Digits & number entry

Primary: Negative sign (–) — makes the next number negative. Different from the subtraction key

shifted (2nd / f / SHIFT layer) (press #6 first): Ans — Ans — recall the last answer into the current expression

#54 E Digits & number entry

Primary: ×10ˣ / EXP / EE — enter a power-of-ten exponent (scientific notation), e.g. 6.02 ×10ˣ 23

Keyboard: e

#55 EXE Operators & execute

Primary: Execute / = — evaluate the line and show the result (EXE / ENTER / =)

Keyboard: Enter =

5 · Worked examples

8 examples, each run on this calculator's keys by the U-45 checker. Follow the chips in order.

Example 5.1 — Rectangular to polar radius

Calculate **tor(3,4)**. Press **Reset** first.

#26 →r → #48 3 → #34 , → #41 4 → #33) → #55 EXE

Display shows 5 — expected 5. ✔ verified on the app.

Example 5.2 — Evaluate the polar slot $r_1 = 2\cos\theta$ at $\theta = 0$

Calculate **R1(0)**. Press **Reset** first.

#24 R_1(→ #51 0 → #33) → #55 EXE

Display shows 2 — expected 2. ✔ verified on the app.

Example 5.3 — Area of the circle $r = 2\cos\theta$ (θ from 0 to π)

Calculate **parea(0,π)**. Press **Reset** first.

#28 ∫A → #51 0 → #34 , → #21 π → #33) → #55 EXE

Display shows 3.141592654 — expected 3.14159265. ✔ verified on the app.

Example 5.4 — Arc length of r = 2cos θ for θ in [0, π]

Calculate **plen(0,π)**. Press **Reset** first.

#29 ∫L → #51 0 → #34 , → #21 π → #33) → #55 EXE

Display shows `6.283185292` — expected 6.28318531. ✔ verified on the app.

Example 5.5 — Parametric $X_1(t) = 3\cos t$ at t = 0

Calculate **X1(0)**. Press **Reset** first.

#25 X_1(→ #51 0 → #33) → #55 EXE

Display shows `3` — expected 3. ✔ verified on the app.

Example 5.6 — Slope of the ellipse X=3cos t, Y=2sin t at t = π/4

Calculate **tslope(π/4)**. Press **Reset** first.

#30 dy/dx → #21 π → #45 ÷ → #41 4 → #33) → #55 EXE

Display shows `-0.6666666667` — expected -0.66666667. ✔ verified on the app.

Example 5.7 — Area enclosed by the ellipse (Green's theorem)

Calculate **garea(0,2π)**. Press **Reset** first.

#6 SHIFT → #28 ∮A → #51 0 → #34 , → #47 2 → #21 π → #33) → #55 EXE

Display shows `18.84955592` — expected 18.84955592. ✔ verified on the app.

Example 5.8 — Polar to rectangular x for r = 2, θ = π/3

Calculate **torx(2,π/3)**. Press **Reset** first.

#27 →x → #47 2 → #34 , → #21 π → #45 ÷ → #48 3 → #33) → #55 EXE

Display shows `1` — expected 1. ✔ verified on the app.

6 · Tricks & hacks

Hack 1: History reuse

Scroll up in the home screen history and press ENTER/EXE to copy an old entry or answer.

Hack 2: Ans chaining

Start an entry with an operator to continue from the previous answer.

Hack 3: Store variables

Use STO▶ / → to store values into letters A–Z, then use them in expressions.

Hack 4: Table of values

Use the TABLE view to scan where a function changes sign before solving.

Hack 5: Window first

If a graph looks empty, adjust the WINDOW / zoom standard.

Hack 6: Edit instead of retyping

After a result, press ◀ or ▶ to bring the last expression back with the cursor in it; change one number and press = again.

Hack 7: Ans chains

Start a line with an operator (e.g. ×) and the calculator inserts Ans automatically — great for compound growth: 1.05 ×, then = = = ...

Hack 8: Exact vs decimal

Natural-display models keep fractions, √ and π exact. Press S⇔D to flip to decimal; press it again to return.

Hack 9: Close brackets for free

Trailing closing brackets can be left off; = closes them for you.

Hack 10: Implicit multiplication

Type 2π or 3(4+1) without ×. Implicit multiplication binds tighter than ÷ on textbook models — use brackets if in doubt.

Hack 11: Store constants

Store a value you reuse (e.g. g = 9.81) with STO and a letter; then type the letter in formulas.

Hack 12: Power chains

x^y chains right to left: 2^3^2 = 2^9 = 512, not 64. Use brackets: (2^3)^2 = 64.

Hack 13: Angles

Check the D / R / G indicator before trig. sin 30 = 0.5 only in degrees.

Hack 14: Memory sums

M+ / M− keep a running total separate from the calculation line; RCL M shows it.

Hack 15: Replay history

Use ▲ ▼ to scroll earlier lines; press = on an old line to re-run it.

Hack 16: Hidden functions on this model

17 extra functions sit on shifted layers. Press the layer key once, then the key: for example #6 then #1 → FORMAT; #6 then #2 → TBLSET; #6 then #4 → CALC; #6 then #5 → TABLE.

Hack 17: Find any key by number

Use Appendix B (function finder): look up the function, read its key number, then find that number on the keypad. Numbers run left→right, top→bottom from #1 to #55.

Hack 18: Type faster on a computer

29 keys on this model have a keyboard shortcut (see chapter 7). Typing digits and operators is much faster than clicking.

Hack 19: Check before you trust

Estimate first (round each number) and compare with the display. If the answer is 10× or 100× off, a decimal point or percent key was missed.

Hack 20: Copy results out

Press Copy in the bar above the calculator to put the display text on the clipboard instead of retyping it.

Hack 21: Colour blind or low vision?

Hi-Contrast colours use black/white/yellow/blue/green/red with black or white text for maximum contrast; key numbers stay on in both colour schemes.

7 · Keyboard shortcuts

With the calculator open on a computer, these physical keys press the numbered keys below (Ctrl/Alt/⌘ combinations are left to the browser). Shifted functions are reached by clicking the shift key first.

Keyboard	#	Key	Does
s	#18	sin	Sine (uses the current angle unit)
c	#19	cos	Cosine (current angle unit)
t	#20	tan	Tangent (current angle unit)
p	#21	π	π (3.14159265…) as an exact constant
q	#22	x^2	Square x^2
^	#23	^	Power x^y / ^ — raise to any power. Chains right-to-left (2^3^2 = 2^9 = 512)
(	#32	(	Opening bracket (
)	#33	)	Closing bracket). Closing brackets at the end of a line may be omitted
,	#34	,	Comma / argument separator (between function arguments, list items or coordinates)
n	#35	ln	Natural logarithm ln
7	#36	7	Digit 7
8	#37	8	Digit 8
9	#38	9	Digit 9
Backspace Delete	#39	DEL	DEL — delete the character at the cursor
Escape	#40	AC	AC — clear all of the current calculation
4	#41	4	Digit 4
5	#42	5	Digit 5
6	#43	6	Digit 6
* x	#44	×	Multiplication ×
/	#45	÷	Division ÷

1	#46	1	Digit 1
2	#47	2	Digit 2
3	#48	3	Digit 3
+	#49	+	Addition +
-	#50	–	Subtraction –
0	#51	0	Digit 0
.	#52	.	Decimal point. Pressing it first gives “0.”; a second point in the same number is ignored
e	#54	E	×10ˣ / EXP / EE — enter a power-of-ten exponent (scientific notation), e.g. 6.02 ×10ˣ 23
Enter =	#55	EXE	Execute / = — evaluate the line and show the result (EXE / ENTER / =)

App shortcuts

“# On / # Off”	show / hide key numbers
“Classic / Hi-Contrast”	switch key colours
“Copy”	copy display text
“Fit / 1:1”	scale to screen / design size
“Reset”	restart this calculator
Jump list	switch to another of the 45 calculators

8 · Common mistakes

Mistake 1: (–) vs –

Use the negative key for negative numbers.

Mistake 2: Radian mode

Graphs of sin x in degrees look flat over −10..10; switch to radians for standard trig windows.

Mistake 3: Implicit multiplication

2/3x may be read as 2/(3x) depending on the model; use brackets.

Mistake 4: Using – instead of (–)

The negative key (–) and the subtraction key are different; "– 5" at the start uses Ans – 5.

Mistake 5: Wrong angle unit

If sin 30 gives −0.988, the calculator is in radians. Switch to degrees in SETUP.

Mistake 6: -3^2 vs $(-3)^2$

$-3^2 = -9$ because the square is applied first. Type $(-3)^2$ to get 9.

Mistake 7: Division precedence

1 ÷ 2π may be read as 1 ÷ (2π) on textbook models. Type 1 ÷ (2 × π) explicitly to be sure.

Mistake 8: Forgetting SHIFT is one-shot

SHIFT applies only to the next key. If the S indicator is still lit, press SHIFT again to cancel.

Mistake 9: Clearing memory by accident

AC clears the line but not memory; CLR (SHIFT 9) can clear memory and variables.

Mistake 10: Forgetting Reset between problems

Old values (memory, stack, open brackets, worksheet values) can leak into the next calculation. Press Reset or clear before starting a new problem.

Mistake 11: Pressing the wrong layer

A shift / 2nd indicator that is still lit will turn the next key into its shifted function. Look at the display indicators before pressing.

Mistake 12: Minus vs negative

The (−) negative key and the − subtraction key are different. Use (−) to start a negative number, − between two numbers.

9 · Practice drills

Set A — Calculation drills (8)

Work each one on calculator #11, then check your keys and the answer in Appendix A. Every answer was produced by this calculator.

1. **tor(3,4)** = _____
2. **R1(0)** = _____
3. **parea(0,π)** = _____
4. **plen(0,π)** = _____
5. **X1(0)** = _____
6. **tslope(π/4)** = _____
7. **garea(0,2π)** = _____
8. **torx(2,π/3)** = _____

Set B — Key-finding drills (62)

Without looking at the table, find each key on the picture. Answers in Appendix A.

1. **Which key is "r(θ)=" (Edit polar and parametric slots)?** _____
2. **Which key is "WINDOW" (WINDOW)?** _____
3. **Which key is "ZOOM" (Zoom menu)?** _____
4. **Which key is "TRACE" (TRACE)?** _____
5. **Which key is "GRAPH" (GRAPH)?** _____
6. **Which key is "SHIFT" (SHIFT / 2nd)?** _____

7. Which key is “MODE” (SETUP)? _____

8. Which key is “TOOLS” (Polar/parametric tools)? _____

9. Which key is “▲” (Cursor ▲)? _____

10. Which key is “▼” (Cursor ▼)? _____

11. Which key is “◀” (Cursor ◀)? _____

12. Which key is “▶” (Cursor ▶)? _____

13. Which key is “HOME” (Calculation screen)? _____

14. Which key is “θ” (Polar angle variable)? _____

15. Which key is “T” (Parameter variable)? _____

16. Which key is “SEL” (Turn the current slot on/off)? _____

17. Which key is “ON” (ON)? _____

18. Which key is “sin” (Sine)? _____

19. Which key is “cos” (Cosine)? _____

20. Which key is “tan” (Tangent)? _____

21. Which key is “π” (π)? _____

22. Which key is “x^2” (Square x^2)? _____

23. Which key is “^” (Power x^y / ^)? _____

24. Which key is “R_1(” (Evaluate polar slot r_1)? _____

25. Which key is “X_1(” (Evaluate X_1)? _____

26. Which key is “→r” (Radius from x,y)? _____

27. Which key is "$\rightarrow x$" (x from r,θ)? _____

28. Which key is "$\int A$" (Polar area of r_1)? _____

29. Which key is "$\int L$" (Polar arc length of r_1)? _____

30. Which key is "dy/dx" (Parametric slope)? _____

31. Which key is "κ" (Curvature of X_1Y_1)? _____

32. Which key is "(" (Opening bracket)? _____

33. Which key is ")" (Closing bracket). Closing brackets at the end of a line may be omitted)? _____

34. Which key is "," (Comma / argument separator)? _____

35. Which key is "ln" (Natural logarithm ln)? _____

36. Which key is "DEL" (DEL)? _____

37. Which key is "AC" (AC)? _____

38. Which key is "×" (Multiplication ×)? _____

39. Which key is "÷" (Division ÷)? _____

40. Which key is "+" (Addition +)? _____

41. Which key is "−" (Subtraction −)? _____

42. Which key is "." (Decimal point. Pressing it first gives "0."; a second point in the same number is ignored)? _____

43. Which key is "(−)" (Negative sign)? _____

44. Which key is "E" (×10^x / EXP / EE)? _____

45. Which key is "EXE" (Execute / =)? _____

46. How do you reach “FORMAT”? _____

47. How do you reach “TBLSET”? _____

48. How do you reach “CALC”? _____

49. How do you reach “TABLE”? _____

50. How do you reach “$\sin^{-1}$”? _____

51. How do you reach “$\cos^{-1}$”? _____

52. How do you reach “$\tan^{-1}$”? _____

53. How do you reach “$\sqrt{\ }$”? _____

54. How do you reach “R_2(”? _____

55. How do you reach “Y_1(”? _____

56. How do you reach “$\rightarrow\theta$”? _____

57. How do you reach “$\rightarrow$y”? _____

58. How do you reach “$\oint$A”? _____

59. How do you reach “$\int$Lt”? _____

60. How do you reach “dy/dθ”? _____

61. How do you reach “ex”? _____

62. How do you reach “Ans”? _____

Appendix A — Drill answers

Set A

1. **5** (display `5`)

   `#26` →r → `#48` 3 → `#34` , → `#41` 4 → `#33`) → `#55` EXE

2. **2** (display `2`)

   `#24` R_1(→ `#51` 0 → `#33`) → `#55` EXE

3. **3.14159265** (display `3.141592654`)

   `#28` ∫A → `#51` 0 → `#34` , → `#21` π → `#33`) → `#55` EXE

4. **6.28318531** (display `6.283185292`)

   `#29` ∫L → `#51` 0 → `#34` , → `#21` π → `#33`) → `#55` EXE

5. **3** (display `3`)

   `#25` X_1(→ `#51` 0 → `#33`) → `#55` EXE

6. **-0.66666667** (display `-0.6666666667`)

   `#30` dy/dx → `#21` π → `#45` ÷ → `#41` 4 → `#33`) → `#55` EXE

7. **18.84955592** (display `18.84955592`)

   `#6` SHIFT → `#28` ∮A → `#51` 0 → `#34` , → `#47` 2 → `#21` π → `#33`) → `#55` EXE

8. **1** (display `1`)

   `#27` →x → `#47` 2 → `#34` , → `#21` π → `#45` ÷ → `#48` 3 → `#33`) → `#55` EXE

Set B

1. #1
2. #2
3. #3
4. #4
5. #5
6. #6
7. #7
8. #8
9. #9
10. #10
11. #11
12. #12
13. #13
14. #14
15. #15
16. #16
17. #17
18. #18
19. #19
20. #20
21. #21
22. #22
23. #23
24. #24
25. #25
26. #26
27. #27
28. #28
29. #29
30. #30
31. #31
32. #32
33. #33
34. #34
35. #35
36. #39
37. #40
38. #44
39. #45
40. #49
41. #50
42. #52
43. #53
44. #54
45. #55
46. #6 SHIFT then #1 r(θ)=
47. #6 SHIFT then #2 WINDOW
48. #6 SHIFT then #4 TRACE
49. #6 SHIFT then #5 GRAPH
50. #6 SHIFT then #18 sin
51. #6 SHIFT then #19 cos
52. #6 SHIFT then #20 tan
53. #6 SHIFT then #22 x^2
54. #6 SHIFT then #24 $R_1($
55. #6 SHIFT then #25 $X_1($
56. #6 SHIFT then #26 →r
57. #6 SHIFT then #27 →x
58. #6 SHIFT then #28 ∫A
59. #6 SHIFT then #29 ∫L
60. #6 SHIFT then #30 dy/dx
61. #6 SHIFT then #35 ln
62. #6 SHIFT then #53 (-)

Appendix B — Function finder (A–Z)

Function	Label	Keys
×10^x / EXP / EE — enter a power-of-ten exponent (scientific notation), e.g. 6.02 ×10^x 23	E	#54
AC — clear all of the current calculation	AC	#40
Addition +	+	#49
Angle from x,y	→θ	#6 → #26
Ans — recall the last answer into the current expression	Ans	#6 → #53
Calculation screen	HOME	#13
Closing bracket). Closing brackets at the end of a line may be omitted	)	#33
Comma / argument separator (between function arguments, list items or coordinates)	,	#34
Cosine (current angle unit)	cos	#19
Cursor ▲ — up / previous history line / previous item	▲	#9
Cursor ▶ — move right in the line	▶	#12
Cursor ▼ — down / next history line / next item	▼	#10
Cursor ◀ — move left in the line	◀	#11
Curvature of X_1Y_1	K	#31
Decimal point. Pressing it first gives "0."; a second point in the same number is ignored	.	#52
DEL — delete the character at the cursor	DEL	#39
Digit 0	0	#51
Digit 1	1	#46
Digit 2	2	#47
Digit 3	3	#48
Digit 4	4	#41
Digit 5	5	#42

Digit 6	6	#43
Digit 7	7	#36
Digit 8	8	#37
Digit 9	9	#38
Division ÷	÷	#45
Edit polar and parametric slots	r(θ)=	#1
Evaluate polar slot r_1	R_1(	#24
Evaluate r_2	R_2(	#6 → #24
Evaluate $X_1(t)$	X_1(	#25
Evaluate $Y_1(t)$	Y_1(	#6 → #25
e^x — exponential	ex	#6 → #35
Execute / = — evaluate the line and show the result (EXE / ENTER / =)	EXE	#55
GRAPH — draw the graph	GRAPH	#5
Inverse cosine $\cos^{-1}$	$\cos^{-1}$	#6 → #19
Inverse sine $\sin^{-1}$	$\sin^{-1}$	#6 → #18
Inverse tangent $\tan^{-1}$	$\tan^{-1}$	#6 → #20
Multiplication ×	×	#44
Natural logarithm ln	ln	#35
Negative sign (−) — makes the next number negative. Different from the subtraction key	(−)	#53
ON — power on	ON	#17
Opening bracket (	(	#32
Parameter variable	T	#15
Parametric arc length	∫Lt	#6 → #29

Parametric enclosed area	**∮A**	**#6 → #28**
Parametric slope	**dy/dx**	**#30**
Polar angle variable	**θ**	**#14**
Polar arc length of r_1	**∫L**	**#29**
Polar area of r_1	**∫A**	**#28**
Polar slope	**dy/dθ**	**#6 → #30**
Polar/parametric tools	**CALC**	**#6 → #4**
Polar/parametric tools	**TOOLS**	**#8**
Power x^y / ^ — raise to any power. Chains right-to-left (2^3^2 = 2^9 = 512)	**^**	**#23**
Radius from x,y	**→r**	**#26**
SETUP — angle unit, number format, display options	**MODE**	**#7**
SHIFT / 2nd — the next key uses its upper (shift-coloured) function	**SHIFT**	**#6**
Sine (uses the current angle unit)	**sin**	**#18**
Square root √	**√**	**#6 → #22**
Square x^2	**x^2**	**#22**
Subtraction −	**−**	**#50**
Table of values	**TABLE**	**#6 → #5**
Tangent (current angle unit)	**tan**	**#20**
TBLSET — table start and step	**TBLSET**	**#6 → #2**
Toggle polar / rectangular grid	**FORMAT**	**#6 → #1**
TRACE — move along a graph	**TRACE**	**#4**
Turn the current slot on/off	**SEL**	**#16**
WINDOW — set the graph view	**WINDOW**	**#2**
x from r,θ	**→x**	**#27**

y from r,θ	**→y**	**#6 → #27**
Zoom menu	**ZOOM**	**#3**
π (3.14159265…) as an exact constant	**π**	**#21**

Appendix C — Glossary

Ans
The result of the last calculation, reusable in the next.

AOS
Algebraic Operating System — operator precedence on a one-line display.

RPN
Reverse Polish Notation — operands first, then the operation; no = key.

Stack
The RPN registers X, Y, Z, T where numbers wait for operations.

TVM
Time Value of Money — n, i, PV, PMT, FV.

Natural display
Shows fractions, roots and powers as printed in textbooks.

Shift / 2nd
Prefix key that selects the function printed above a key.

Alpha
Prefix key that types the letter printed beside a key.

Radian
Angle unit; 2π rad = 360°.

Grad
Angle unit; 400 grad = 360°.

Engineering notation
Scientific notation with exponents that are multiples of 3.

Mantissa
The digits of a number in scientific notation.

Memory (M)

An independent register for running totals.

Mark-up

Increase over cost; margin is profit as a percent of selling price.

Grand total (GT)

Sum of all totals since the last clear.

nCr / nPr

Combinations / permutations of n items taken r at a time.

Factorial

n! = 1×2×…×n.

Reciprocal

1/x.

Standard deviation

Spread of data around the mean (σ population, s sample).

Key number

The small badge 1…N on each key in U-45 CALCULATORS, used throughout this manual.

Appendix D — Model notes & limits

Polar & parametric grapher. Angles inside the graph slots are always radians; θ and t are separate variables.

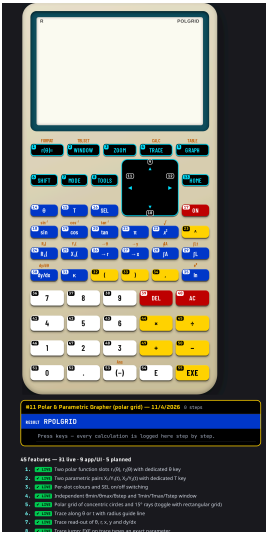
Areas use ½∫r²dθ (polar) and ½∮(x dy − y dx) (parametric). Arc length integrates √(r² + (dr/dθ)²) or √(x′² + y′²) with Simpson's rule (800 panels).

Planned (not yet working)

- Graph image export (PNG)
- Shareable function list link
- Dark screen theme
- Speech read-out of values
- Exam mode

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Appendix E — Quick-reference card



#1 r(θ)= ⇧ FORMAT	#2 WINDOW ⇧ TBLSET	#3 ZOOM
#4 TRACE ⇧ CALC	#5 GRAPH ⇧ TABLE	#6 SHIFT
#7 MODE	#8 TOOLS	#9 ▲
#10 ▼	#11 ◀	#12 ▶
#13 HOME	#14 θ	#15 T
#16 SEL	#17 ON	#18 sin ⇧ $\sin^{-1}$
#19 cos ⇧ $\cos^{-1}$	#20 tan ⇧ $\tan^{-1}$	#21 π
#22 x^2 ⇧ √	#23 ^	#24 R_1(⇧ R_2(
#25 X_1(⇧ Y_1(	#26 →r ⇧ →θ	#27 →x ⇧ →y
#28 ∫A ⇧ ∮A	#29 ∫L ⇧ ∫Lt	#30 dy/dx ⇧ dy/dθ
#31 K	#32 (	#33)
#34 ,	#35 ln ⇧ e^x	#36 7
#37 8	#38 9	#39 DEL
#40 AC	#41 4	#42 5
#43 6	#44 ×	#45 ÷
#46 1	#47 2	#48 3
#49 +	#50 −	#51 0

#52 .	**#53** (−) ⇧ Ans	**#54** E
#55 EXE		

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